

Applied Computational Methods for Social Sciences

TECNOLÓGICO DE MONTERREY

Spring 2026

bgpopescu.net/teaching/text_analysis

Course Information

Course Number: P1012

Course Title: Text Analysis

Time: Tue/Fri: 13:10pm-14:50pm

Location: Aulas VII 103

Instructor Details

Instructor: Bogdan Popescu

Office: xx

E-Mail: xx

Office Hours: xx

Course Description

This is an introductory course on using Python, a free and versatile programming language, to analyze text data. The course begins by building foundational programming skills in Python, including working with variables, loops, conditional statements, and data structures like lists, dictionaries, and arrays. In the second part, students will explore Python's powerful tools for analyzing text data, such as social media posts, email correspondence, customer reviews, and political debates. The course also includes an introduction to ChatGPT and prompt engineering, teaching students how to leverage large language models for advanced text analysis tasks.

Through hands-on exercises, lectures, and projects, students will learn to process, clean, and analyze textual data systematically and reproducibly. By the end of the course, participants will be equipped with the skills to derive meaningful insights from textual information and integrate generative AI tools like ChatGPT into their workflows.

Note: You should have a computer either Windows or Mac with at least 8GB of RAM memory.

Learning Outcomes

Upon successful completion of this course the students will be able to:

- Write Python programs to perform tasks such as loops, conditional statements, and function definitions.
- Employ quantitative techniques to process and analyze textual data.
- Utilize Python libraries like Numpy, Pandas, and NLTK for text manipulation and analysis.
- Apply advanced methods such as sentiment analysis, topic modeling, word embeddings, and supervised learning to text data.
- Use ChatGPT and prompt engineering to enhance text analysis tasks, including summarization, classification, and generating structured outputs.
- Create and publish a professional website showcasing their portfolio and analytical capabilities.

Textbook and/or Resource Materials

No textbook required

Assessment

Assessment methods:

You will be graded on four problem sets during the semester (each 12.5% of your grade) and a final report and presentation (each 25% of your grade).

- 4 problem sets: 50% of the final grade (12.5% each)
- final presentation: 25% of the final grade
- final project report: 25% of the final grade

Problem Sets

1.Initial Individual Submission: This component contributes 50% of the overall grade for the problem set. When you complete the problem set independently and submit it to the instructor, your grade for this component will be calculated based on the quality of your independent work. This grade will be weighted at 50% of the total assignment grade.

2.Final Submission After Group Consultation: This component also contributes 50% of the overall grade for the problem sets. After discussing the problem set with your group members and documenting the correct answers, you will submit this revised version individually. Note: no group submission is permitted. Each one of you has to submit the second attempt of the assignment individually. Your grade for this component will be based on the quality of your final submission after group consultation. This grade will also be weighted at 50% of the total assignment grade.

Presentation

Each student will deliver a 10-15 minute presentation on a topic assigned in advance. Presentations should include an overview of the:

- question
- description of the selected text analysis method
- description of the dataset used
- description of how the data was validated
- critical analysis

Please deliver your presentations in a quarto presentation and use relevant visuals. Practice beforehand to stay within the time limit and maintain a confident, professional tone. Be prepared to answer 2-3 questions from peers or the instructor during and after the presentation. Remember to cite your sources and avoid reading verbatim from slides or notes.

In addition to summarizing the key arguments or findings, your presentation should include critical analysis of the material. Highlight what the author does not address, the limitations of their research, or potential problems in their analysis or methodology. Think about how the research could be improved, expanded, or connected to broader themes discussed in class, and incorporate these insights into your presentation.

Final Project

Students will undertake a text analysis project emphasizing the practical application of text analysis techniques using the python programming language. This project offers an opportunity to showcase the acquired skills in manipulating text data and conducting meaningful analyses. Participants are encouraged to choose a topic of interest or relevance, utilizing datasets provided in the course or exploring new ones. The project entails a few steps:

- choosing a topic that involves text data
- acquire data either from the course materials or from external sources

Students have to submit a two-page report in Quarto and give a 20-minute and give an in-class presentation in Quarto.

Grading Criteria for the project:

- Relevance
- Methodology
- Analysis
- Presentation
- Code Quality

Here are examples of projects that previous students created:

- [Topical Similarity in the US Presidential Speeches](#)
- [Framing Latin America in UN Speeches](#)
- [Agreesion in Political Speeches.](#)

Attendance

Students are required to attend classes following the University's policies. Students with more than three unexcused absences are assumed to have withdrawn from the course. Students with a justified reason not to attend class have to send me an email explaining why they cannot attend ahead of class.

Academic Honesty

As stated in the university catalog, any student who commits an act of academic dishonesty will receive a failing grade on the work in which the dishonesty occurred. In addition, acts of academic dishonesty, irrespective of the weight of the assignment, may result in the student receiving a failing grade in the course. Instances of academic dishonesty will be reported to the Dean of Academic Affairs. A student reported twice for academic dishonesty is subject to summary dismissal from the University. In such a case, the Academic Council will then make a recommendation to the President, who will make the final decision.

Students with Learning Difficulties and other Disabilities

The University does not discriminate based on disability. Students with approved accommodations must inform their professors at the beginning of the term. Please see the website for the complete policy.

Required Books

General Books on Python

McKinney, Wes, 2022. *Python for Data Analysis*. O'Reilly. <https://wesmckinney.com/book/>.

Text Analysis

Hovy, Dirk. 2020. *Text Analysis in Python for Social Scientists*. Cambridge University Press. <https://www.cambridge.org/core/elements/abs/text-analysis-in-python-for-social-scientists/BFAB0A3604C7E29F6198EA2>

Bengfort, Benjamin, Bilbro, Rebecca, and Tony Ojeda. 2018. *Applied Text Analysis with Python*. O'Reilly Media, Inc. <https://www.oreilly.com/library/view/applied-text-analysis/9781491963036/>

Hvitfeldt, Emil and Silge, Julia. 2022. *Supervised Machine Learning for Text Analysis in R*. CRC Press. <https://smltar.com>

Grimmer, Justin, Brandon M. Stewart, and Margaret E. Roberts. *Text As Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press, 2022.

Silge, Julia, and David Robinson. *Text Mining with R: A Tidy Approach*. First edition. O'Reilly, 2017.

1 Installing R and Python

Lecture 1

- Introduction and course overview
- Installing R, R Studio, and Anaconda
- Introduction to Python programming and its applications in text analysis
- Overview of text data (e.g., reviews, political debates, translations)

[Download the Slides](#)

Lecture 2

- Quarto files: notebooks vs. presentations
- The Python environment

[Download the Slides](#)

Lecture 3

- How to Make Slides in Quarto

[Download the Slides](#)

Reading

McKinney, Wes, 2022. *Python for Data Analysis*. O'Reilly. <https://wesmckinney.com/book/>. C1-2

2 Variables, Strings, Loops

Lecture 3

- Variables
- Strings
- Combining strings
- Numbers
- Booleans
- Conditionals: `if` and `elif`

[Download the Slides](#)

Lecture 4

- Comparison Operators
- Error Handling
- Lists
- Loops
- Breaks

[Download the Slides](#)

Assignment 1

[See the Assignment](#)

3 Lists, Tuples, Loops

Lecture 5: Operations with Lists and Tuples

- Lists
- List Operations
- List Slicing

- Tuples

[Download the Slides](#)

Lecture 5: While Loops and Functions

- Comparison Operators
- Error Handling
- Lists
- Loops
- Breaks

[Download the Slides](#)

Reading

McKinney, Wes, 2022. *Python for Data Analysis*. O'Reilly. <https://wesmckinney.com/book/>. C3

4 Dictionaries, Pandas, Dataframes

Lecture 7: Dictionaries and Sets

- dictionaries
- extracting elements from dictionary
- sorting dictionary keys
- nesting dictionaries
- dictionaries within dictionaries
- sets and operations with sets

[Download the Slides](#)

Lecture 8: Pandas Data Wrangling

- Dataframes
- Dataframe Methods
- Missing Data
- Assigning Values

[Download the Slides](#)

Lecture: Intro to Numpy

[Download the Slides](#)

5 Data Wrangling

Lecture 9: Pandas Data Wrangling

- Setting Indexes
- Changing, ordering column names
- Identifying and Removing duplicates

- Filtering Data
- Aggregating Data
- Modifying Data Structures

[Download the Slides](#)

Lecture 10: Merging Data

- Aggregating by Group
- Inner, Outer Joins
- Left, Right Joins

[Download the Slides](#)

Assignment 2

[See the Assignment](#)

6 Data Visualization in R

Lecture 11: Data Visualization in R - 1

[Download the Slides](#)

Video Lecture 11: Data Visualization in R - 1

[Watch the Video](#)

Lecture 12: Data Visualization in R - 2

[Download the Slides](#)

Video Lecture 12: Data Visualization in R - 2

[Watch the Video](#)

Lecture 13: Data Visualization in R - 3

[Download the Slides](#)

Video Lecture 13: Data Visualization in R - 3

[Watch the Video](#)

7 Text as Data and ChatGPT

Lecture 13: Text as Data

- Latent Concepts

- Dictionaries, Supervised learning, text scaling, topic models, word embeddings
- Workflows of Text Analysis
- Documents, features,
- Bag-of-Words and Document-feature matrix
- Text Cleaning in Python

[Download the Slides](#)

Lecture 14: Intro to ChatGPT for Text Analysis

- Prompt engineering
- Personas, delimiters, structured outputs, Programmatic consumptions
- Chain of Thought
- ChatGPT Api, pricing

[Download the Slides](#)

8 Applied ChatGPT and Dictionary Methods

Lecture 13: Applied ChatGPT for Text Analysis

- Text Summarization, Text Classification
- Sentiment Analysis, Emotion Analysis
- Topic Modeling, Topics using Zero-Shot Learning
- Language, Grammar and Spelling

[Download the Slides](#)

Lecture 14: Dictionary Methods

- Counting words
- Examples of Dictionaries
- Validation of Dictionary Methods
- Accuracy, Sensitivity, Specificity, Naive Guess

[Download the Slides](#)

9 Similarity and Word Clouds

Lecture 14: Similarity

- Inner Product
- Frequency words, Euclidean distance
- Cosine Similarity

[Download the Slides](#)

Lecture 14: Similarity and Word Clouds

- Similarity: Levenshtein distance

- Cosine similarity
- TF-IDF, Bag-of-Words
- Word Clouds, Fightin' Words

[Download the Slides](#)

Assignment 3

[See the Assignment](#)

10 Supervised Learning

Lecture: Designing Projects using Text Analysis

[Download the Slides](#)

Lecture 19: Language Complexity

- Flesch Reading Ease
- Kincaid score

[Download the Slides](#)

Lecture 20: Supervised Learning

- Naive Bayes
- Language Models
- Validating Supervised Learning, K-Fold Validation

[Download the Slides](#)

11 Unsupervised Learning

Lecture 19: Unsupervised Learning: Topic Models

- Topic Models
- Latent Dirichlet Allocation
- Semantic Coherence

[Download the Slides](#)

Lecture 20: Unsupervised Learning: Word Embeddings

- Word Embeddings
- Distributional Semantics, Topic Distributions
- BERTopic

[Download the Slides](#)

Assignment 4

[See the Assignment](#)

12 Student Presentations

Lecture: Student presentations

- Blumenau, Jack E; Hargrave, Lotte, 2022. [No Longer Conforming to Stereotypes? Gender, Political Style, and Parliamentary Debate in the UK](#). British Journal of Political Science - **Wendy and Nico**
- Blaydes Lisa; Grimmer, Justin; McQueen, Alison, 2018. [Mirrors for Princes and Sultans: Advice on the Art of Governance in the Medieval Christian and Islamic Worlds](#). Journal of Politics - **Harman**
- Bellodi, Luca, 2022. [A Dynamic Measure of Bureaucratic Reputation: New Data for New Theory](#). American Journal of Political Science - **Kayla**
- Motolinia, Lucia, 2020. [Electoral Accountability and Particularistic Legislation: Evidence from an Electoral Reform in Mexico](#). American Journal of Political Science - **Catalina**
- Riccardo Di Leo, Chen Zeng, Elias Dinas, and Reda Tamtam, 2025. [Mapping \(A\)Ideology: A Taxonomy of European Parties Using Generative LLMs as Zero-Shot Learners](#). Political Analysis - **Martina**

Lecture: Student Proposals

1. Question
2. Data sources
3. Proposed Text Analysis functions
4. Expected policy implications

13 Sentiment Analysis & GitHub Websites

Lecture 21: Sentiment Analysis

- Lexicon-based approaches, machine learning
- VADER sentiment analysis, transformers
- Zero-shot classification

[Download the Slides](#)

Lecture 22: Making a Quarto Website

- Making a website on Quarto
- Storing a website on GitHub

- Structuring a professional website for Data Analytics Jobs
- Website Troubleshooting

[Download the Slides](#)